QC.5) Same as QC.4, but include Cohen-Coon and do the experiment for: $Q_2 = 0$

Q_2 = step function

Manufacturer $\begin{cases} K_c = 70 \\ \tau_I = 50\,s \\ \tau_D = 7\,s \end{cases}$

IMC $\begin{cases} K_c = 74.71 \\ \tau_I = 76.2\,s \end{cases} \rightarrow \tau_D = \dfrac{\tau_p \theta_p}{2\tau_p + \theta_p} = 2.944\,s$

Cohen-Coon: $K_c = \dfrac{\tau_p}{K_p \theta_p}\left(\dfrac{4}{3} + \dfrac{\theta_p}{4\tau_p}\right) = 53.59$

$\tau_I = \theta_p\left(\dfrac{32 + 6r}{13 + 8r}\right) = 14.53\,s$

$\left(r = \dfrac{\theta_p}{\tau_p} = 0.037974\right)$

$\tau_D = \theta\left(\dfrac{4}{11 + 2r}\right) = 2.763\,s$